

Database Design

Data-va epdi store pannradhu

Database Design na yenna?

System-ku thevaiyaana data-va tables-a, relationships-a organize pannni store pannra plan.

Steps

- **Step 1:** Entities identify pannunga (User, Order, Product).
- **Step 2:** Ovvoru entity-ku attributes ezhudhunga.
- **Step 3:** Primary key select pannunga.
- **Step 4:** Relationships define pannunga.
- **Step 5:** ER diagram varainga.
- **Step 6:** Normalization pannunga.
- **Step 7:** Tables create panna SQL ezhudhunga.

Relationship Types

Type	Example
One-to-One	User - Profile.
One-to-Many	Customer - Orders.
Many-to-Many	Students - Courses (junction table use pannunga).

Normalization

Form	Rule
1NF	Ovvoru cell-la oru value mattum, repeating groups illa.
2NF	1NF + partial dependency illa.
3NF	2NF + transitive dependency illa.

Sample Table Design

Column	Type	Key	Note
user_id	INT	PK	Auto increment
name	VARCHAR(100)		Required
email	VARCHAR(150)	Unique	Login-ku

Column	Type	Key	Note
created_at	DATETIME		Default now

Checklist

- Ovvoru table-ukkum primary key irukka?
- Foreign keys sariya irukka?
- Duplicate data illaya?
- Search-ku indexes thevaiyaa?
- Sensitive data (password) encrypt / hash panniyaachaa?